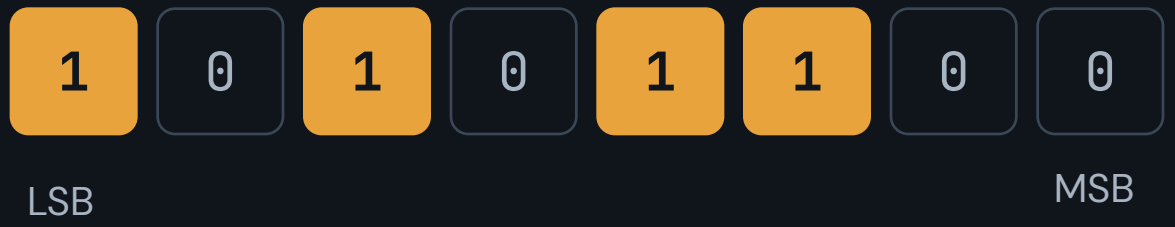
Stored functions

Argument row registers, the instruction word, and the memtrix that shows where everything lives



One row is 8 cells, one byte. This one holds 0x35, with bit 0 on the left.

Functions move out of the assembler and into program memory

BEFORE: INLINE MACROS

Every use pastes the same NANDs into main. The chip never knows a function existed.

220

words for adding 4 numbers with three inline copies of add8

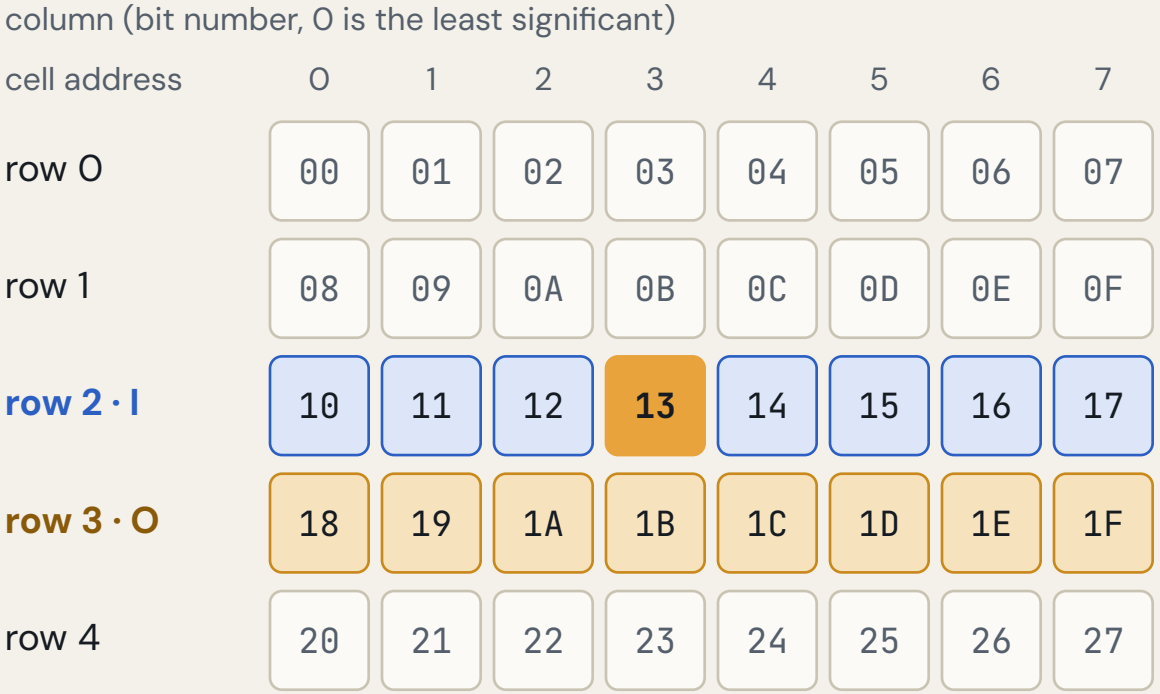
AFTER: STORED BODIES

The body is stored once. Each use is a single CALL word.

75

words for the same 3-variable-add: 3 calls in main plus a 72-word body, at one extra cycle per call

Memory is a grid of rows and columns



A row is one byte

Eight cells, one bit each. Arguments are always whole rows.

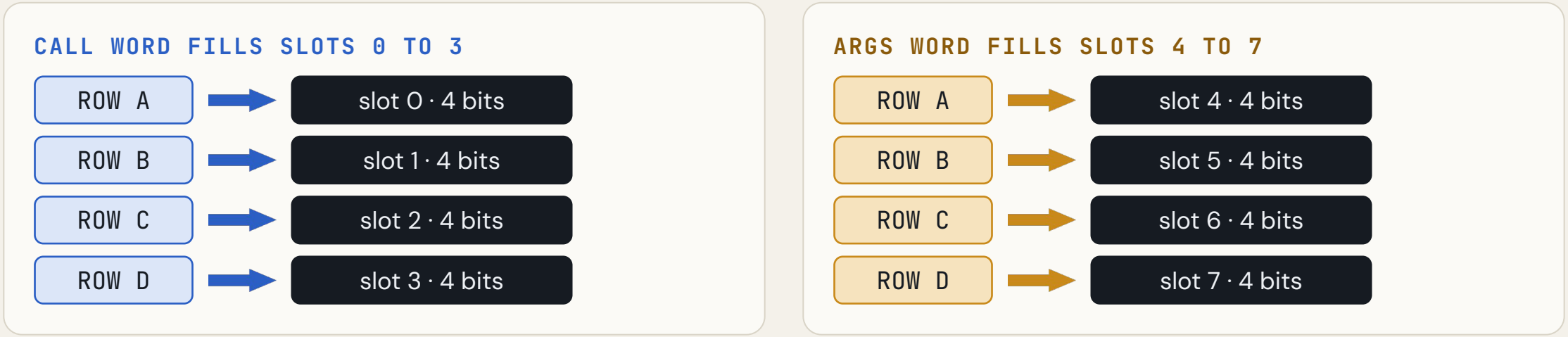
A column is one bit

Column 0 is the least significant bit.

0x13 = 2 × 8 + 3

row 2, column 3: the highlighted cell

Row registers hold the rows of one call



32 flip-flops

Eight registers of 4 bits, loaded in parallel from the word

Link register

8 bits. Saves PC + 1 at CALL, restores it at RET

Read through the window

Every body operand picks one register with a slot mux

RAM address 0x13

0, then the row, then the column



if slot 0 holds row 2

0x13

if slot 0 holds row 3

0x1B

Same body word, different cell on each call. The word never changes; only the register does.

One 28-bit word, read two ways

NAND word

- ADDRESS A and ADDRESS B: the two input cells
- DST: the cell that receives the result
- In a body these can be window addresses

CALL word

- ROW A to ROW D: the rows for slots 0 to 3, 4 bits each
- LOC: where the body starts, the jump target
- The same 24 bits, reused, so the word stays 28 bits

What the flags tell the control unit

Word	OP	RAM write	Next PC	Row registers	Link register
HALT	000	no	stay	unchanged	unchanged
NAND	001	yes	PC + 1	unchanged	unchanged
NAND + RET	010	yes	link	unchanged	unchanged
CALL	011	no	LOC	load slots 0–3	save PC + 1
ARGS	100	no	PC + 1	load slots 4–7	unchanged

HALT is 000, so erased program RAM stops the machine by itself.

ARGS goes right before its CALL, and never appears inside a body.

The RAM write is on for

`OP = 001 or 010`

R	0x00	0x01	0x02	0x03	0x04	0x05	0x06	0x07	0x08	0x09	0x0A	0x0B	0x0C	0x0D	0x0E	0x0F	R
0	MAIN															HALT	1
2	BODIES																3
4																	5
6																	7
8																	9
10																	11
12																	13
14																	15
16																	17
18																	19
20																	21
22																	23
24																	25
26																	27
28																	29
30																	31

R	PROG
1	0x00
3	0x10
5	0x20
7	0x30
9	0x40
11	0x50
13	0x60
15	0x70
17	0x80
19	0x90
21	0xA0
23	0xB0
25	0xC0
27	0xD0
29	0xE0
31	0xF0

R	0x00	0x01	0x02	0x03	0x04	0x05	0x06	0x07	0x08	0x09	0x0A	0x0B	0x0C	0x0D	0x0E	0x0F	R
0	INIT	NOT	AND	OR	XOR	SHL	SHR	ADD	X	X	X	X	X	X	X	END	1
2	NOT'S								X	X	X	X	X	X	X	X	3
4	AND'S																5
6	OR'S																7
8	XOR'S																9
10																	11
12									X	X	X	X	X	X	X	X	13
14	SHL'S															X	15
16	SHR'S															X	17
18	ADD'S																19
20																	21
22																	23
24																	25
26											X	X	X	X	X	X	27
28	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	29
30	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	31

av	0x00	0x01	0x02	0x03	0x04	0x05	0x06	0x07	0x08	0x09	0x0A	0x0B	0x0C	0x0D	0x0E	0x0F	av
0	CONSTANTS				I BITS				O BITS								1
2	PROTECTED VARIABLES																3
4																	5
6																	7
8																	9
10																	11
12																	13
14																	15
16	CALLER SCRATCH																17
18																	19
20	FUNCTION SCRATCH																21
22																	23
24	ARGUMENT WINDOW																25
26																	27
28																	29
30																	31

R	MEMTRIX
1	0x00
3	0x10
5	0x20
7	0x30
9	0x40
11	0x50
13	0x60
15	0x70
17	0x80
19	0x90
21	0xA0
23	0xB0
25	0xC0
27	0xD0
29	0xE0
31	0xF0

0x00	0x01	0x02	0x03	0x04	0x05	0x06	0x07	0x08	0x09	0x0A	0x0B	0x0C	0x0D	0x0E	0x0F	cd
0	1			CIN	X	X	X	POS	ZERO	EQ	GRT	OVF	X	X	X	1
A								B								3
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	5
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	7
NOT A								AND A B								9
OR A B								XOR A B								11
ADD A B								SHR A								13
SHL A								X	X	X	X	X	X	X	X	15
?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	17
?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	19
?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	21
?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	23
ARG A								ARG B								25
ARG C								ARG D								27
X								X								29
X								X								31

R	0x00	0x01	0x02	0x03	0x04	0x05	0x06	0x07	0x08	0x09	0x0A	0x0B	0x0C	0x0D	0x0E	0x0F	R
R	DATA																R

BIT COUNT ->	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
NAND WORD ->	ADDRESS A								ADDRESS B								DST								OP		
CALL WORD ->	ROW A				ROW B				ROW C				ROW D				LOC								OP		

Seven regions in two memories

Region	Memory	Rows	Addresses	Holds
MAIN	program	0–15	0x00–0x7F	the program that runs
BODIES	program	16–31	0x80–0xFF	function bodies, stored in order
CONST	data	0	0x00–0x01	the constants 0 and 1
PROTECTED VARIABLES	data	2–15	0x10–0x7F	the caller's rows; arguments live here
CALLER SCRATCH	data	16–19	0x80–0x9F	temporaries of main's own code
FUNCTION SCRATCH	data	20–23	0xA0–0xBF	temporaries of bodies, shared
PARAMETER WINDOW	data	24–31	0xC0–0xFF	8 slots, with no RAM behind them

The dark cells in row 0 (0x02 to 0x04) are not used in the memtrix (for now).

Running it, one word per cycle

Cycle	Word	What happens
1	INIT	NAND(0x00, 0x00) writes 0x01, so the constant 1 is set.
2	I1 (CALL)	Slot 0 takes row 2 (I), slot 1 takes row 3 (O), the link register takes 0x02 and the PC jumps to 0x80. Nothing is written to RAM.
3 to 10	NOT8 body	NAND(ARG A[i], ARG A[i]) writes ARG B[i], for i = 0 to 7. The last word carries RET.
11	END	RET returned to 0x02, which is END, so the machine halts.

Cycles to run

1 + 1 + 8 = 10

The same eight body words would serve any other pair of rows. Only the CALL word changes.

How ARG A and ARG B reach A and B

ARG A · 0xC0–0xC7
slot 0 holds row 2

→ **A · 0x10–0x17**
row 2, the input

ARG B · 0xC8–0xCF
slot 1 holds row 3

→ **B · 0x18–0x1F**
row 3, the output

```
0x80 NAND(0xC0, 0xC0) → 0xC8 reads 0x10, writes 0x18 (bit 0)
0x87 NAND(0xC7, 0xC7) → 0xCF reads 0x17, writes 0x1F (bit 7)
```

A = 0x35	1	0	1	0	1	1	0	0
B = 0xCA	0	1	0	1	0	0	1	1

Bit 0 is on the left. The value 0x35 is picked for illustration; the memtrix does not fix one.

Rules the design relies on

Bodies are leaves

A stored body never contains a CALL: one link register, one set of rows.

Arguments are whole rows

An argument starts on an 8-aligned address, 0x10 to 0x78.

Two scratch regions

Main and bodies each get their own temporaries.

Scratch does not last

What must outlive a call lives in a variable row.

Empty words read as HALT

An all-zero word would overwrite the constant 0, so empty words must stop the machine.

The window has no RAM

Operands from 0xC0 to 0xFF always go through a row register.